

Traffic Light PCB Design

eSim / KiCad PCB Design Project – Task 7

1. Project Overview

This project is a Traffic Light PCB designed using the eSim/KiCad based PCB design workflow. The design demonstrates schematic creation, component footprint assignment, PCB layout, routing, design rule checking, and 3D visualization.

2. Circuit Description

The circuit implements a simple traffic light system using LEDs to represent the traffic signals. The PCB contains the required electronic components and connections arranged on a compact two-layer board.

3. Schematic Design

The schematic was created in the eSim schematic environment. Components were connected according to the intended traffic light circuit and the schematic was checked before generating the PCB layout.

4. Footprint Assignment

Appropriate PCB footprints were assigned to the components so that the schematic could be transferred to the PCB layout. Component placement was selected to keep the board organized and easy to route.

5. PCB Layout and Routing

The PCB uses two copper layers, satisfying the minimum layer requirement. Components were placed on the board according to functional and routing considerations. Tracks were routed between the components while keeping the layout clean and avoiding unnecessary crossings.

6. Design Rule Check

A Design Rule Check (DRC) was performed on the completed PCB. Critical layout issues were checked and the final design was verified before submission.

7. 3D Visualization

The completed PCB was inspected using the KiCad/eSim 3D Viewer. This was used to verify the board outline, component placement, and overall physical appearance of the PCB.

8. Project Files

File	Purpose
trafficlight.kicad_sch	Schematic
trafficlight.kicad_pcb	PCB layout
trafficlight.kicad_pro	Project configuration
trafficlight.kicad_prl	Project settings
trafficlight_sch.pdf	Schematic documentation

9. Conclusion

The Traffic Light PCB project demonstrates the complete PCB design workflow from schematic capture to PCB layout, routing, DRC verification, and 3D inspection. The final project is organized for submission as an eSim/KiCad PCB design project.

Submission: The complete project folder, report, and supporting documents should be uploaded to the private GitHub fork as required by the assignment.